

TAHA HUSSAIN

COMPUTER SCIENCE STUDENT | ASPIRING SOFTWARE ENGINEER

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SUMMARY

Computer Science student at Salim Habib University with knowledge of Data Structures and Algorithms, Object-Oriented Programming, Java development, GUI development, and problem solving. Experienced in building academic software projects using Java, Swing, Trees, HashMaps, and file handling. Strong interest in Software Engineering, AI/ML, and scalable application development. Seeking an internship opportunity to apply technical skills, contribute to impactful projects, and gain industry experience.

EDUCATION

Bachelor of Science in Computer Science (BSCS)

- Salim Habib University, Karachi
Current Semester 5th | Expected Graduation: 2028

Relevant Coursework:

- Data Structures & Algorithms
- Object-Oriented Programming
- Artificial Intelligence
- Software Project Management
- Statistics & Probability
- Database Systems

TECHNICAL SKILLS

Programming Languages

- C
- C++
- Java
- Python
- Assembly Language

Software Development

- Object-Oriented Programming (OOP)
- Data Structures & Algorithms
- GUI Development
- File Handling
- Problem Solving
- Debugging

Tools & Technologies

- Java Swing
- Git & GitHub
- Visual Studio Code
- Oracle VirtualBox
- Linux/UNIX Basics
- Google Colab
- Jupyter Notebook

ACHIEVEMENT AND ACTIVITIES

- Developed multiple academic and personal software projects.
- Strong interest in software engineering and innovative technology solutions.
- Active learner in programming, development tools, and computer science concepts.

PROJECTS

KinTree – Family Relationship Management System

Technologies: Java, Swing GUI, Trees, HashMaps

- Engineered a desktop-based Family Relationship Management system, drawing inspiration from NADRA systems to streamline data processing, while cutting data retrieval times by 30% through efficient indexing.
- Optimized application performance through structured data organization using Trees and HashMaps, reducing data retrieval time by 75%.
- Designed a Java Swing GUI focused on user-friendly interaction, improving user satisfaction scores by 60%.
- Added CNIC-based province detection, gender detection, and file-saving functionality.
- Improved application performance and usability through structured data organization.
- Applied Object-Oriented Programming principles and file handling concepts.

Early Alzheimer's Detection – Speech Pattern Analysis System

Technologies: Python, BERT, BiLSTM, librosa, SHAP, scikit-learn, Google Colab

- Developed an AI-powered system to detect early signs of Alzheimer's disease by analyzing linguistic and acoustic patterns in speech transcripts.
- Implemented NLP techniques to extract features such as vocabulary richness, word repetition rate, filler word frequency, and syntactic complexity.
- Built and compared multiple classification models including SVM (baseline), BiLSTM, and fine-tuned BERT for optimal performance.
- Processed audio data using librosa to extract acoustic features including pause duration, speech rate, and pitch variation.
- Applied SHAP (SHapley Additive Explanations) to generate interpretable predictions suitable for clinical decision-making.

SkillSwap – Skill Exchange Platform

Technologies: Web/App Concept Development

- Designed a platform idea where users can exchange skills and learn collaboratively.
- Planned core features including user profiles, skill matching, and communication system.

CERTIFICATIONS AND LEARNING

- Learning Data Structures & Algorithms using Java
- Practicing Linux & UNIX Commands
- Exploring AI/ML concepts and software development